

(1001CJA102120047)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

FULL SYLLABUS

24-01-2021

**JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)****ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

| SECTION-I  | Q. | 1    | 2            | 3                | 4    | 5            | 6    | 7 | 8   | 9       | 10      |
|------------|----|------|--------------|------------------|------|--------------|------|---|-----|---------|---------|
|            | A. | D    | A            | B                | B    | A            | B    | C | B,C | A,B,C,D | A,B,C,D |
|            | Q. | 11   | 12           |                  |      |              |      |   |     |         |         |
| SECTION-II | A. | A,B  | A,C          |                  |      |              |      |   |     |         |         |
|            | Q. | 1    | 2            | 3                | 4    | 5            | 6    |   |     |         |         |
|            | A. | 3.00 | 0.66 to 0.67 | 286.66 to 286.67 | 3.00 | 0.56 to 0.57 | 2.18 |   |     |         |         |

**PART-2 : CHEMISTRY**

| SECTION-I  | Q. | 1     | 2       | 3    | 4    | 5    | 6                | 7       | 8     | 9       | 10  |
|------------|----|-------|---------|------|------|------|------------------|---------|-------|---------|-----|
|            | A. | C     | A       | C    | A    | D    | A                | A,B,C,D | A,B,C | A,B,C,D | C,D |
|            | Q. | 11    | 12      |      |      |      |                  |         |       |         |     |
| SECTION-II | A. | A,B,D | A,B,C,D |      |      |      |                  |         |       |         |     |
|            | Q. | 1     | 2       | 3    | 4    | 5    | 6                |         |       |         |     |
|            | A. | 3.00  | 5.00    | 1.91 | 4.00 | 4.00 | -38.85 to -38.95 |         |       |         |     |

**PART-3 : MATHEMATICS**

| SECTION-I  | Q. | 1     | 2       | 3            | 4     | 5    | 6     | 7   | 8   | 9     | 10  |
|------------|----|-------|---------|--------------|-------|------|-------|-----|-----|-------|-----|
|            | A. | B     | A       | A            | A     | C    | A     | A,D | B,C | A,B,C | A,D |
|            | Q. | 11    | 12      |              |       |      |       |     |     |       |     |
| SECTION-II | A. | A,C,D | A,B,C,D |              |       |      |       |     |     |       |     |
|            | Q. | 1     | 2       | 3            | 4     | 5    | 6     |     |     |       |     |
|            | A. | 9.00  | 1.00    | 1.87 or 1.88 | 15.00 | 0.75 | 12.56 |     |     |       |     |

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## CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

FULL SYLLABUS

24-01-2021

### JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

#### ANSWER KEY

#### PAPER-2

##### PART-1 : PHYSICS

| SECTION-I   | Q. | 1            | 2     | 3     | 4            | 5     | 6            |
|-------------|----|--------------|-------|-------|--------------|-------|--------------|
|             | A. | A,C          | B,C,D | B,C   | A,C          | A,B   | A,C          |
| SECTION-II  | Q. | 1            | 2     | 3     | 4            | 5     | 6            |
|             | A. | 0.76 to 0.77 | 57.82 | 13.12 | 5.64 to 5.66 | 61.20 | 5.80 to 6.00 |
| SECTION-III | Q. | 1            | 2     | 3     | 4            | 5     | 6            |
|             | A. | 3            | 3     | 8     | 0            | 1     | 5            |

##### PART-2 : CHEMISTRY

| SECTION-I   | Q. | 1     | 2     | 3       | 4     | 5       | 6     |
|-------------|----|-------|-------|---------|-------|---------|-------|
|             | A. | B,C   | A,B,C | A,B,C,D | A,B,D | A,B,C,D | A,C   |
| SECTION-II  | Q. | 1     | 2     | 3       | 4     | 5       | 6     |
|             | A. | -3.00 | 7.00  | -105.50 | 4.00  | 4.00    | 94.00 |
| SECTION-III | Q. | 1     | 2     | 3       | 4     | 5       | 6     |
|             | A. | 2     | 4     | 3       | 9     | 4       | 3     |

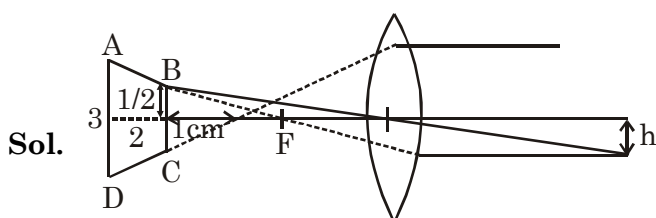
##### PART-3 : MATHEMATICS

| SECTION-I   | Q. | 1    | 2    | 3    | 4    | 5            | 6    |
|-------------|----|------|------|------|------|--------------|------|
|             | A. | A,C  | B,D  | A,C  | B,C  | A,D          | A,B  |
| SECTION-II  | Q. | 1    | 2    | 3    | 4    | 5            | 6    |
|             | A. | 4.80 | 0.50 | 0.84 | 0.50 | 0.24 or 0.25 | 1.50 |
| SECTION-III | Q. | 1    | 2    | 3    | 4    | 5            | 6    |
|             | A. | 1    | 2    | 1    | 4    | 2            | 3    |

**JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)****PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (D)**

**Sol.** The initial velocity of smoke particles die out quickly due to air resistance. The Buoyant force is vertically up. So they move in upward direction.

But train moves forwards, so smoke trail is at an acute angle behind the train.

2. **Ans. (A)**

$$\frac{h}{1} = m_1 = \frac{f}{-x_0}$$

$$\frac{h}{3} = m_2 = \frac{f}{-(x_0 + 2)}$$

$$\frac{f}{3x_0} = \frac{f}{x_0 + 2}$$

$$\Rightarrow x_0 + 2 = 3x_0$$

$$x_0 = 1 \text{ cm}$$

$$x_0 x_I = -f^2$$

$$1 \times x_I = +f^2 \Rightarrow x_I = f^2$$

$$x_I = \frac{f^2}{3}$$

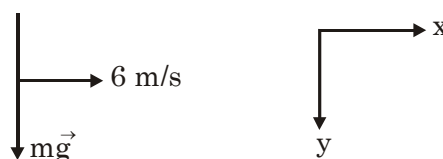
$$x_I - I'_I = \frac{2}{3} f^2 = h = f$$

$$f = 1.5 \text{ cm}$$

$$h = 1.5 \text{ cm}$$

3. **Ans. (B)****Sol.**  $mg = C \times 8$ 

$$-C \times \vec{v}_{m\omega}$$



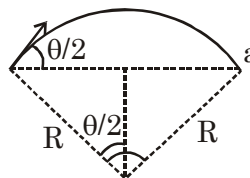
$$-C \times \vec{v}_{m\omega} = mg$$

$\vec{v}_{m\omega}$  should be vertically down.

$$\Rightarrow \vec{v}_{m\omega} = 8 \text{ m/s } \hat{i}$$

$$\vec{v}_m - \vec{v}_\omega = 8 \hat{i}$$

$$\vec{v}_m = 8 \hat{i} + 6 \hat{j}$$

4. **Ans. (B)****Sol.**

$$\theta = \omega t = \frac{qBt}{m}$$

$$\sin \frac{\theta}{2} = \frac{a}{2R}$$

$$\Rightarrow \frac{\theta}{2} = \sin^{-1} \left( \frac{aqB}{2p_0} \right)$$

$$t = \frac{2m}{qB} \sin^{-1} \left( \frac{aqB}{2p_0} \right) = \frac{2m}{qB} \times \frac{\pi}{6}$$

$$= \frac{12}{\pi \times 1/5} \times \frac{\pi}{6} = 20 \text{ sec}$$

5. Ans. (A)

$$\begin{aligned} \text{Sol. } P &= i^2 R = \frac{v_0^2}{2(x_0^2 + (R_1 + R_2)^2)} \times R_2 \\ &= \frac{900 \times 8}{2\left(\left(\frac{1}{5000 \times 2\pi} \times 10^6\right)^2 + 20^2\right)} \\ &= \frac{3600}{1000 + 400} = \frac{3600}{1400} = 2.57 \text{ W} \end{aligned}$$

6. Ans. (B)

$$\begin{aligned} \text{Sol. } \frac{df}{f^2} &= \frac{du}{u^2} + \frac{dv}{v^2}, \quad du = dv \\ \frac{1}{v} - \frac{1}{u} &= \frac{1}{f} \Rightarrow \frac{1}{u^2} + \frac{1}{v^2} - \frac{2}{uv} = \frac{1}{f^2} \\ \frac{df}{f^2} &= du \left( \frac{1}{f^2} + \frac{2}{uv} \right) = du \left( \frac{1}{f^2} + \frac{2}{u} \left( \frac{1}{f} + \frac{1}{u} \right) \right) \\ df &= du \left( 1 + \frac{2f}{u} + \frac{2f^2}{u^2} \right) \\ \Rightarrow df &\text{ is minimum when} \\ 0 &= -\frac{2f}{u^2} - \frac{4f^2}{u^3} \\ u &= -2f \\ (\text{for } u &= \infty, 2^{\text{nd}} \text{ derivative is zero so it is not a minimum.}) \end{aligned}$$

7. Ans. (C)

$$\begin{aligned} \text{Sol. } \frac{h}{\mu v_{\text{rel}}} &= \frac{5}{8} \frac{h}{m_1 v_1} = \frac{5}{2} \frac{h}{m_2 v_2} \\ m_2 v_2 &= \pm 4 m_1 v_1 \\ \text{Taking +ve signs} \\ \frac{m_1 m_2}{m_1 + m_2} (v_1 - v_2) &= \frac{8}{5} m_1 v_1 \\ m_2 v_1 - m_2 v_2 &= \frac{8}{5} m_1 v_1 + \frac{8}{5} m_2 v_1 \\ \frac{3}{5} m_2 v_1 &= -\frac{7}{5} m_1 v_1 \\ \text{takign } v_{\text{rel}} &= v_2 - v_1 \end{aligned}$$

$$\Rightarrow \frac{m_2(v_2 - v_1)}{m_1 + m_2} = \frac{8}{5} v_1$$

$$m_2 v_2 - m_2 v_1 = \frac{8}{5} m_1 v_1 + \frac{8}{5} m_2 v_1$$

$$\frac{3}{5} m_2 v_2 = \frac{13}{5} m_2 v_1$$

$$v_2 = \frac{13}{3} v_1$$

$$m_2 v_2 = 4 m_1 v_1$$

$$\frac{v_2}{v_1} = \frac{4 m_1}{m_2} = \frac{13}{3}$$

$$\frac{m_1}{m_2} = \frac{13}{12} \text{ not possible as } m_1 < m_2$$

Taking -ve sign

$$m_2 v_2 = -4 m_1 v_1$$

$$m_2 v_1 - m_2 v_2 = +\frac{8}{5} m_1 v_1 + \frac{8}{5} m_2 v_1$$

$$-\frac{3}{5} m_2 v_1 = \frac{3}{5} m_2 v_2$$

$$v_1 = -v_2$$

or

$$m_2 v_2 - m_2 v_1 = \frac{8}{5} m_1 v_1 + \frac{8}{5} m_2 v_1$$

$$m_2 v_2 - \frac{8}{5} m_1 v_1 = \frac{13}{5} m_2 v_1$$

$$\frac{7}{5} m_2 v_2 = \frac{13}{5} m_2 v_1$$

$$\frac{v_2}{v_1} = \frac{13}{7} \times \text{not possible as } \frac{m_2}{m_1} = -ve$$

8. Ans. (B,C)

Sol. Applying Bernoulli's theorem from the frame of car.

$$P_{\text{in}} + 0 = P_{\text{out}} + \frac{1}{2} \rho v^2$$

$$(P_{\text{in}} - P_{\text{out}}) \times (0.4)^2 = \sigma \times 1.6 \times t$$

$$\Rightarrow \frac{1}{2} \rho v^2 \times 0.16 = 4 \times 10^7 \times 1.6t$$

$$\Rightarrow \frac{1}{2} \times \frac{1.25 \times 40 \times 40 \times 0.16}{1.60 \times 4 \times 10^7} = t$$

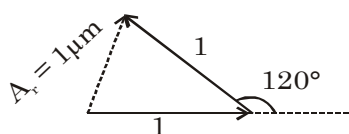
$$t = 2.5 \mu\text{m}$$

9. Ans. (A,B,C,D)

$$\text{Sol. } S_1 = 1 \mu\text{m} \sin\left(\frac{\pi}{4}x - 100\pi t - \frac{\pi}{3}\right)$$

$$= 1 \mu\text{m} \sin\left(\frac{\pi}{4}x - 100\pi t + \pi - \frac{\pi}{3}\right)$$

$$= 1 \mu\text{m} \sin\left(\frac{\pi}{4}x - 100\pi t + \frac{2\pi}{3}\right)$$



10. Ans. (A,B,C,D)

$$\text{Sol. } Q = nC_v \times 15 = 600$$

$$\Rightarrow nC_v = 40$$

$$Q = nC_p \times 10 = 600$$

$$nC_p = 60$$

$$n(C_p - C_v) = 20$$

$$n \times \frac{25}{3} = 20$$

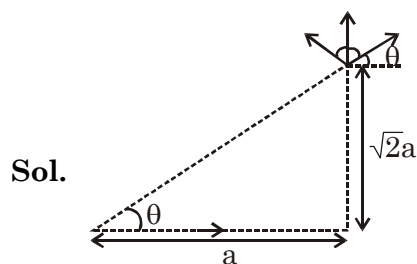
$$n = 2.4$$

$$\frac{C_p}{C_v} = 1.5$$

$$C_v = 2R = \frac{n_1 \times \frac{5}{2}R + n_2 \times \frac{3}{2}R}{n_1 + n_2}$$

$$\Rightarrow n_1 = n_2$$

11. Ans. (A,B)

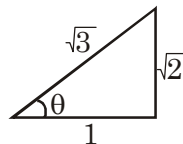


Sol.

$$\frac{2kpcos^2\theta}{r^3} = \frac{kpsin^2\theta}{r^3}$$

$$\tan\theta = \sqrt{2}$$

$$\frac{kpcos\theta sin\theta}{\sqrt{3}a^3} = \frac{kq}{2a^2}$$



12. Ans. (A,C)

$$\text{Sol. } R_1 + R_2 = 4R_0$$

$$\Rightarrow \frac{R_1}{R_0} + \frac{R_2}{R_0} = 4 \Rightarrow x + y = 4$$

$$\Rightarrow \frac{\frac{R_1 R_2}{R_1 + R_2}}{R_0} = \frac{40}{60} = \frac{2}{3}$$

$$\Rightarrow R_1 R_2 = \frac{2}{3} 4R_0^2$$

$$\Rightarrow xy = \frac{2}{3} \times 4 = \frac{8}{3}$$

$$x + \frac{8}{3x} = 4$$

$$3x^2 - 12x + 8 = 0$$

$$x = \frac{12 \pm \sqrt{144 - 4 \times 8 \times 3}}{6} = \frac{R_1}{R_0} = \frac{x}{100 - x}$$

$$0.84 = \frac{x}{100 - x} \Rightarrow x \approx 46 \text{ cm}$$

## SECTION-II

1. Ans. 3.00

$$\text{Sol. } P_{\alpha\text{-decay}} = 0.25 \alpha \lambda_\alpha = \frac{\ln 2}{T_\alpha}$$

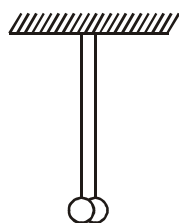
$$P_\beta = 0.75 \alpha \lambda_\beta = \frac{\ln 2}{T_\beta}$$

$$\Rightarrow \frac{1}{3} = \frac{T_\beta}{T_\alpha}$$

$$\Rightarrow \frac{T_\alpha}{T_\beta} = 3$$

2. Ans. 0.66 to 0.67

Sol.



$$T = 2\pi\sqrt{\frac{\ell}{g}} = \text{Same for both.}$$

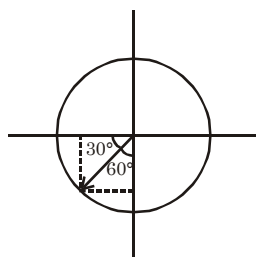
$\Rightarrow$  collision at mean position of the bobs.

$\Rightarrow$  velocities are exchanged

$\Rightarrow$  Amplitude are exchanged

$$A_{\text{left bob}} = 6^\circ$$

$$T_1 = \frac{T}{4}$$



$$\omega T_2 = \frac{\pi}{6}$$

$$T_2 = \frac{T}{12}$$

$$\Rightarrow t = \frac{T}{4} + \frac{T}{12} = \frac{4T}{12} = \frac{T}{3} = 0.66 \text{ sec}$$

3. Ans. 286.66 to 286.67

Sol. Apply the condition for constructive interference twice:

$$\frac{(4\lambda_1)}{d} = \frac{x}{D} \text{ and } \frac{(6\lambda_2)}{d} = \frac{x}{d},$$

So that  $4\lambda_1 = 6\lambda_2$ .

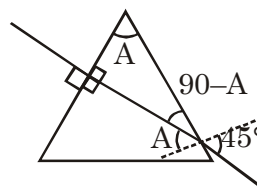
If  $\lambda_1 = 430 \text{ nm}$ , then  $\lambda_2 = 287 \text{ nm}$ .

4. Ans. 3.00

$$\text{Sol. } v_1 = -L_1 \frac{dI_1}{dt} - M \frac{dI_2}{dt}$$

$$= -(0.050)(120) - (0.015)(-200) = -3V$$

5. Ans. 0.56 to 0.57



Sol.

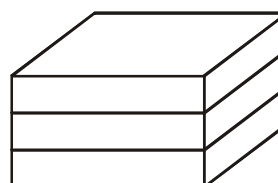
$$n \sin A = 1 \sin 45^\circ$$

$$n = \frac{C}{v} = \frac{3 \times 10^8}{2.4 \times 10^8} = \frac{5}{4}$$

$$\sin A = \frac{2\sqrt{2}}{5} = \frac{2.82}{5}$$

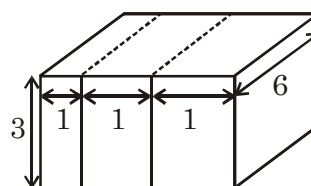
$$= 0.56 \text{ to } 0.57$$

6. Ans. 2.18



Sol.

$$C_1 = \frac{\epsilon_0 \times 18}{\frac{1}{1} + \frac{1}{2} + \frac{1}{3}} = \frac{\epsilon_0 \times 18}{\frac{6+3+2}{6}} = \epsilon_0 \times \frac{108}{11}$$



$$C_2 = \frac{1 \times \epsilon_0 \times 1 \times 6}{3} + \frac{2 \times \epsilon_0 \times 1 \times 6}{3} + \frac{3 \times \epsilon_0 \times 1 \times 6}{3}$$

$$= \epsilon_0 \times 12$$

$$\Rightarrow \frac{C_2 - C_1}{\epsilon_0} = 12 \left[ 1 - \frac{9}{11} \right] = \frac{24}{11} = 2.18$$

**PART-2 : CHEMISTRY**

**SOLUTION**

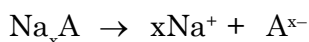
**SECTION-I**

1. **Ans.(C)**

2. **Ans.(A)**

**Sol.** At isoelectric point Amino acid has least viscosity. Therefore option (A) is correct.

3. **Ans.(C)**



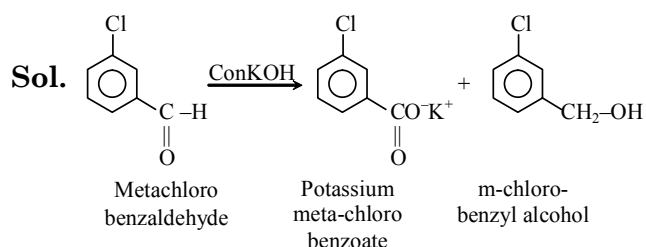
$$\Delta T_f = i k_f m$$

$$3.255 = (x + 1) \times 1.86 \times 0.35$$

$$x = 4$$

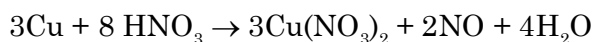
4. **Ans.(A)**

5. **Ans.(D)**



It is an example of cannizzaro reaction. Therefore option (D) is correct.

6. **Ans.(A)**



$$\frac{95.25}{63.5} = 1.5\text{mol} \quad \frac{8}{3} \times \frac{3}{2} = 4\text{mol}$$

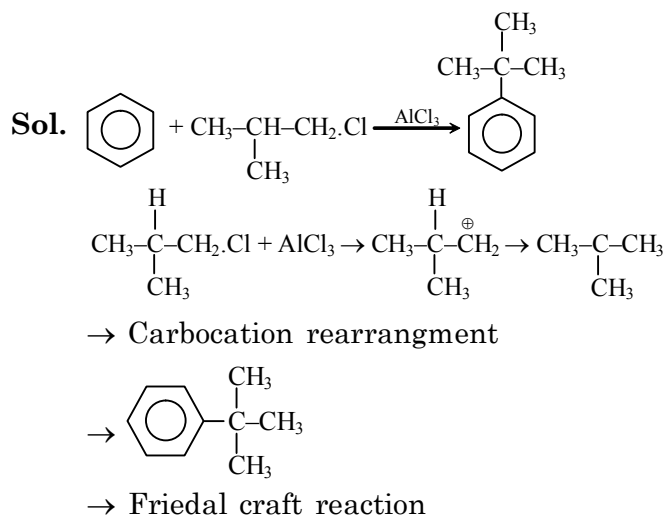
$$= M \times V(l)$$

$$= 1 \times V(l)$$

$$\therefore V = 4.0 \text{ litre.}$$

7. **Ans.(A,B,C,D)**

8. **Ans.(A,B,C)**



9. **Ans.(A,B,C,D)**

Initial mass of  $\text{H}_2\text{SO}_4$  in the solution = 1260  
 $\times 0.40 = 504 \text{ gm } \text{H}_2\text{SO}_4$

Say Wgm  $\text{H}_2\text{SO}_4$  consumed.

$$\frac{(504 - w)}{\left(1260 - w + \frac{18}{98}w\right)} \times 100 = 28$$

$$w = 196$$

$$n_{\text{H}_2\text{SO}_4} = \frac{196}{98} = 2$$

2 mol  $\text{H}_2\text{SO}_4$  consumed  $\equiv 2F$  charge passed  
 $= 2 \times 96500 = 1.93 \times 10^5 \text{ coulomb.}$

10. **Ans.(C,D)**

11. **Ans.(A,B,D)**

12. **Ans.(A,B,C,D)**

Angular function  $\propto \cos \theta \Rightarrow$  orbital  $p_z$

Power of  $\sigma$  is 1  $\Rightarrow l = 1 \Rightarrow$  p-orbital

$$n - l - 1 = 1 \Rightarrow n = 3$$

$\therefore$  orbital is  $3p_z$

Angular node =  $l = 1$

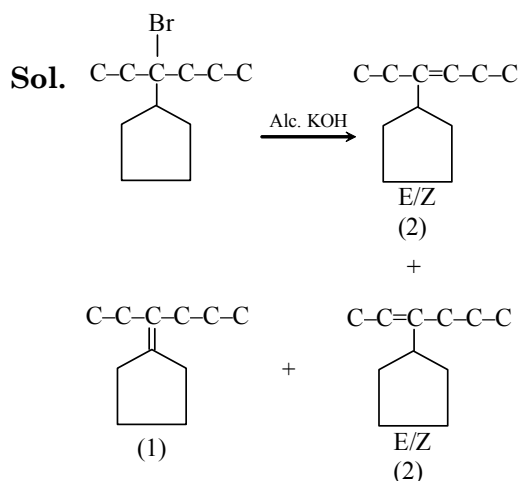
Radial node =  $n - l - 1 = 1$

XY - plane is nodal plane

**SECTION-II**

1. **Ans.(3.00)**

2. **Ans.(5.00)**



Total five isomer

3. **Ans.(1.91)**

$$\ln 2 = \frac{E_a}{R} \left[ \frac{1}{280} - \frac{1}{290} \right]$$

$$\ln \mu = \frac{E_a}{R} \left[ \frac{1}{290} - \frac{1}{300} \right]$$

$$\frac{\ln \mu}{\ln 2} = \frac{280 \times 290}{290 \times 300}$$

$$\mu = 1.91$$

4. **Ans.(4.00)**

5. **Ans.(4.00)**

6. **Ans.(-38.85 to -38.95)**

$$\Delta G = \Delta H - \Delta(T.S)$$

$$\text{or } \Delta G = n.C_{p,m}(T_2 - T_1) - (T_2.S_2 - T_1.S_1) \dots\dots(1)$$

$$\text{Now, } \Delta S = S_{H_2(g, 600K, 2bar)} - S_{H_2(g, 300K, 1bar)}$$

$$= n.C_{p,m} \ln \frac{T_2}{T_1} - nR \ln \frac{P_2}{P_1}$$

$$\text{or, } S_{H_2(g, 600K, 2bar)} - 130.0$$

$$= 1 \times 28.314 \times \ln \frac{600}{300} - 1 \times 8.314 \times \ln \frac{2}{1}$$

$$\therefore S_{H_2(g, 600K, 2bar)} = 144.0 \text{ J / K - mol}$$

$$\text{Eq(1) : } \Delta G = 1 \times 28.314 (600 - 300)$$

$$- (600 \times 144 - 300 \times 130)$$

$$= -38905.8 \text{ J/mol} = -38.9058 \text{ kJ/mol.}$$

## PART-3 : MATHEMATICS

## SOLUTION

### SECTION-I

1. **Ans. (B)**

$$a_1 a_2 a_3 \dots a_i a_{i+1} a_{i+2} \dots a_n$$

$$a_{i+1} = \frac{a_i + a_{i+2}}{2}$$

$$a_i + a_{i+2} = 2a_{i+1}$$

$$S_n = \sum_{i=1}^n \frac{a_i a_{i+1} a_{i+2}}{2a_{i+1}} = \sum_{i=1}^n \frac{a_i a_{i+2}}{2}$$

$$S_n = \frac{1}{2} \sum_{i=1}^n [(5 + (i-1)3)(5 + (i+1)3)]$$

$$S_n = \frac{1}{2} \sum_{i=1}^n (3i+2)(8+3i)$$

$$S_n = \frac{1}{2} \sum_{i=1}^n (9i^2 + 30i + 16)$$

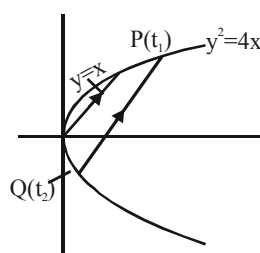
$$S_n = \frac{1}{2} \left[ 9 \cdot \sum_{i=1}^n i^2 + 30 \sum_{i=1}^n i + \sum_{i=1}^n 16 \right]$$

$$S_n = \frac{1}{2} \left[ 9 \cdot \frac{n(n+1)(2n+1)}{6} + 30 \cdot \frac{n(n+1)}{2} + 16n \right]$$

$$S_{10} = \frac{1}{2} \left[ \frac{9 \times 10 \times 11 \times 21}{6} + \frac{30 \times 10 \times 11}{2} + 16 \times 10 \right]$$

$$S_{10} = \frac{1}{2} [3465 + 1650 + 160] = 2637.50$$

2. **Ans. (A)**



Let  $P(t_1^2, 2t_1)$  and  $Q(t_2^2, 2t_2)$

$$\text{slope of chord } PQ = \frac{2}{t_1 + t_2}$$

$$\frac{2}{t_1 + t_2} = 1 \therefore t_1 + t_2 = 2 \dots(1)$$

Equation of normal at P

$$y = -t_1 x + 2t_1 + t_1^3$$

equation of normal at Q

$$y = -t_2 x + 2t_2 + t_2^3$$

point of intersection of normals at P & Q

$$x = t_1^2 + t_2^2 + t_1 t_2 + 2 \dots(3)$$

$$y = -t_1 t_2 (t_1 + t_2) \dots(4)$$

using (1)  $y = -2t_1 t_2 \dots(2)$

$$x = (t_1 + t_2)^2 - 2t_1 t_2 + t_1 t_2 + 2$$

$$x = (2)^2 - t_1 t_2 + 2 = 6 - t_1 t_2 = 6 - \left( -\frac{y}{2} \right)$$

$$x = 6 + \frac{y}{2}$$

$$2x = 12 + y$$

$$2x - y - 12 = 0$$

it is satisfied by point (7,2)

3. **Ans. (A)**

$$\lim_{x \rightarrow 0^+} \left[ \frac{x}{\pi} \right] = \lim_{h \rightarrow 0} \left[ \frac{h}{\pi} \right] = 0$$

$$\lim_{x \rightarrow 0^+} \frac{\sin^2 x}{\left[ \frac{x}{\pi} \right] + \frac{x^2}{\pi^2}} \times \frac{\sin(\sin x) - \sin x}{ax^5 + bx^3 + c} = -\frac{\pi^2}{12}$$

$$x = 0 + h$$

$$\lim_{h \rightarrow 0} \frac{\sin^2 h}{\frac{h^2}{\pi^2}} \times \frac{2 \cos\left(\frac{\sinh+h}{2}\right) \sin\left(\frac{\sinh-h}{2}\right)}{ah^5 + bh^3 + c} = -\frac{\pi^2}{12}$$

$$\lim_{h \rightarrow 0} \frac{\pi^2 \cdot \sin^2 h}{h^2} \times 2 \cos\left(\frac{\sinh+h}{2}\right)$$

$$\frac{\sin\left(\frac{\sinh-h}{2}\right)}{\frac{\sinh-h}{2}} \times \frac{\sinh-h}{ah^5 + bh^3 + c} = -\frac{\pi^2}{12}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\sinh-h}{ah^5 + bh^3 + c} = -\frac{1}{12}$$

$$\lim_{h \rightarrow 0} \frac{\sinh-h}{h^3(ah^2+b)} = -\frac{1}{12} \quad (c=0)$$

$$-\frac{1}{6(b)} = -\frac{1}{12} \therefore b = 2$$

$$a \in \mathbb{R}, b = 2, c = 0$$

$$b + c = 2 + 0 = 2$$

4. **Ans. (A)**

$$(x-1)(x^2-2)(x^3-3) \dots (x^{12}-12)$$

Highest power of x in product

$$\text{is } 1 + 2 + 3 \dots + 12 = \frac{12 \times 13}{2} = 78$$

$$(1) (x^{70} + \dots) (x^1 - 1) (x^7 - 7) = 7x^{70}$$

$$(2) (x^{70} + \dots) (x^2 - 2) (x^6 - 6) = 12x^{70}$$

$$(3) (x^{70} + \dots) (x^3 - 3) (x^5 - 5) = 15x^{70}$$

$$(4) (x^{70} + \dots) (x - 1) (x^2 - 2) (x^5 - 5) = -10x^{70}$$

$$(5) (x^{70} + \dots) (x - 1) (x^3 - 3) (x^4 - 4) = -12x^{70}$$

$$(6) (x^{70} + \dots) (x^8 - 8) = -8x^{70}$$

$$\text{coefficient of } x^{70} = 7 + 12 + 15 - 10 - 12 - 8 = 4$$

5. **Ans. (C)**

Using cross product of three vectors

$$\hat{i} \times [(\vec{a} - \hat{j}) \times \hat{i}]$$

$$\Rightarrow (\hat{i} \cdot \hat{i})(\vec{a} - \hat{j}) - (\hat{i} \cdot (\vec{a} - \hat{j}))\hat{i}$$

$$\Rightarrow \vec{a} - \hat{j} - (\vec{a} \cdot \hat{i})\hat{i}$$

$$\text{Let } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\Rightarrow \vec{a} - \hat{j} - (a_1\hat{i}) \quad \dots(1)$$

similarly :

$$\hat{j} \times [(\vec{c} - \hat{k}) \times \hat{j}]$$

$$\Rightarrow \vec{a} - \hat{k} - a_2\hat{j} \quad \dots(2)$$

$$\text{And : } \hat{k} \times [(\vec{a} - \hat{i}) \times \hat{k}]$$

$$\vec{a} - \hat{i} - a_3\hat{k} \quad \dots(3)$$

Adding (1) (2) and (3)

$$3\vec{a} - (\hat{i} + \hat{j} + \hat{k}) - (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) = 0$$

$$3\vec{a} - (\hat{i} + \hat{j} + \hat{k}) - \vec{a} = 0$$

$$\vec{a} = \frac{\hat{i} + \hat{j} + \hat{k}}{2}$$

$$\therefore \vec{a} \cdot \vec{b} = \left( \frac{\hat{i} + \hat{j} + \hat{k}}{2} \right) \cdot 2(\hat{i} + \hat{j} + \hat{k}) = 3$$

6. **Ans. (A)**

$$2|z^2 + |z|^2| + 3|z^2 - |z|^2| = 6|z|$$

$$2|z(z + \bar{z})| + 3|z(z - \bar{z})| = 6|z|$$

$$2|z| |z + \bar{z}| + 3|z| |z - \bar{z}| = 6|z|$$

$$2|z + \bar{z}| + 3|z - \bar{z}| = 6$$

$$4|x| + 6|y| = 6$$

$$\frac{|x|}{\frac{3}{2}} + \frac{|y|}{2} = 1$$

$$\text{Area} = 4 \times \frac{1}{2} \times \frac{3}{2} \times 1$$

$$\Rightarrow 3 \text{ unit}^2$$

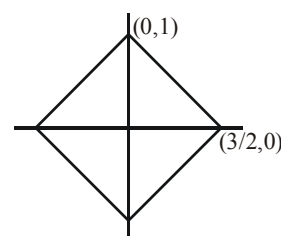
7. **Ans. (A,D)**

$$\frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{4x^2}{1+x^2}$$

$$\text{IF : } e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = (1+x^2)$$

solution of equation :

$$y(1+x^2) = \int \frac{4x^2}{(1+x^2)} (1+x^2) dx$$



$$y(1+x^2) = \frac{4x^3}{3} + c$$

since curve passes through origin

$$x=0, y=0$$

$$0=0+c \therefore c=0$$

$$y(1+x^2) = \frac{4x^3}{3}$$

$$f(x) = y = \frac{4x^3}{3(1+x^2)}$$

$$\frac{dy}{dx} = \frac{(1+x^2)(12x^2) - 4x^3(2x)}{3(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{12x^2 + 4x^4}{3(1+x^2)^2} = \frac{4x^2(3+x^2)}{3(1+x^2)^2}$$

$$\frac{dy}{dx} > 0 \quad \forall x - \{0\} \quad \text{and Also } y = \frac{4x^3}{3(1+x^2)}$$

$$\text{As } x \rightarrow \infty, y \rightarrow \infty$$

$$x \rightarrow -\infty, y \rightarrow -\infty$$

$$A = \frac{2}{3} - \int_0^1 \left( \frac{4x^3}{3(1+x^2)} dx \right)$$

$$A = \frac{2}{3} - \frac{2}{3} \int_0^1 \frac{2x \cdot x^2}{1+x^2} dx$$

$$\int_0^1 \frac{2x \cdot x^2}{1+x^2} dx$$

$$1+x^2 = t$$

$$2x dx = dt$$

$$\int_1^2 \frac{(t-1)}{t} dt$$

$$[t - \ln t]_1^2$$

$$(2 - \ln 2) - (1 - 0) = 1 - \ln 2$$

$$A = \frac{2}{3} - \frac{2}{3} [1 - \ln 2]$$

$$A = \frac{2}{3} \ln 2$$

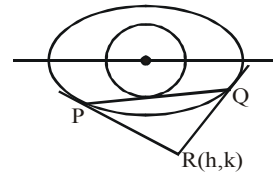
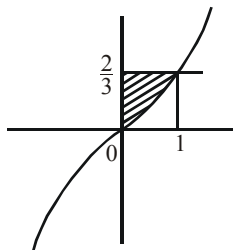
8. **Ans. (B,C)**

Let the equation of tangent to circle

$$x^2 + y^2 = 1 \text{ be } x \cos \theta + y \sin \theta = 1 \quad \dots(1)$$

This tangent intersection ellipse in P and Q.

Let point of intersection of tangent at P and Q be R(h,k)



PQ will be chord of contact w.r.t R(h,k)

$$xh + 2yk = 4 \quad \dots(2)$$

comparing (1) and (2)

$$\frac{\cos \theta}{h} = \frac{\sin \theta}{2k} = \frac{1}{4}$$

$$\cos \theta = \frac{h}{4}, \sin \theta = \frac{2k}{4}$$

$$\therefore h^2 + 4k^2 = 16 \text{ or } \frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$\therefore e = \sqrt{1 - \frac{4}{16}}$$

$$e = \frac{\sqrt{3}}{2}$$

9. **Ans. (A,B,C)**

Let  $\angle B = \theta$

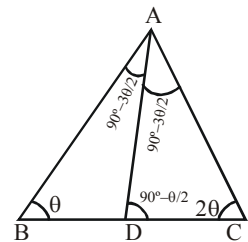
$$\therefore \angle C = 2\angle B = 2\theta$$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\angle A + \theta + 2\theta = 180^\circ$$

$$\angle A = 180^\circ - 3\theta$$

$$\frac{\angle A}{2} = 90^\circ - \frac{3\theta}{2}$$



$$\text{Now in } \triangle ABD \quad \frac{AD}{\sin \theta} = \frac{BD}{\sin \left( 90^\circ - \frac{3\theta}{2} \right)}$$

$$AD = \frac{BD \sin \theta}{\cos \frac{3\theta}{2}} \quad \dots(1)$$

$$\text{In } \triangle ADC \quad \frac{AD}{\sin 2\theta} = \frac{AC}{\sin \left( 90^\circ - \frac{\theta}{2} \right)}$$

$$\text{since } AC = BD \quad AD = \frac{BD \sin 2\theta}{\cos \frac{\theta}{2}} \quad \dots(2)$$

From (1) and (2)

$$\frac{\sin \theta}{\cos \frac{3\theta}{2}} = \frac{\sin 2\theta}{\cos \frac{\theta}{2}}$$

$$\sin \theta \cos \frac{\theta}{2} = \sin 2\theta \cdot \cos \frac{3\theta}{2}$$

$$\sin \theta \cos \frac{\theta}{2} = 2 \sin \theta \cos \theta \cdot \cos \frac{3\theta}{2}$$

$$\cos \frac{\theta}{2} = \cos \frac{5\theta}{2} + \cos \frac{\theta}{2}$$

$$\cos \frac{5\pi}{2} = 0 \quad \therefore \frac{5\theta}{2} = 90^\circ$$

$$\theta = 36^\circ$$

Hence

$$\angle B = 36^\circ, \angle C = 72^\circ$$

$$\angle A = 180^\circ - 3\theta = 180^\circ - 108^\circ = 72^\circ$$

$$\therefore \angle A = \angle C$$

Now check the options

10. Ans. (A,D)

$E_1$  : Person owns mercedes independently

$$P(E_1) = \frac{3}{10}; P(\bar{E}_1) = \frac{7}{10}$$

$E_2$  : Person owns audi independently

$$P(E_2) = \frac{4}{10}; P(\bar{E}_2) = \frac{6}{10}$$

D : Event that person keeps driver

$$P(D) = P(E_1 \cap \bar{E}_2)P\left(\frac{D}{E_1 \cap \bar{E}_2}\right) +$$

$$P(\bar{E}_1 \cap E_2)P\left(\frac{D}{\bar{E}_1 \cap E_2}\right) +$$

$$P(E_1 \cap E_2)P\left(\frac{D}{E_1 \cap E_2}\right)$$

$$P(D) = \frac{3}{10} \times \frac{6}{10} \times \frac{6}{10} + \frac{7}{10} \times \frac{4}{10} \times \frac{7}{10} + \frac{3}{10} \times \frac{4}{10} \times \frac{9}{10}$$

$$P(D) = \frac{412}{1000}$$

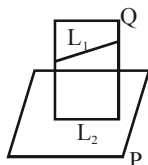
$$P\left(\frac{E_2}{D}\right) = \frac{\frac{7 \times 4 \times 7}{1000} + \frac{3 \times 4 \times 9}{1000}}{\frac{412}{1000}}$$

$$= \frac{304}{412} = \frac{76}{103}$$

11. Ans. (A,C,D)

$$\text{Let } L_1 : \frac{x}{2} = \frac{y-1}{2} = \frac{z-1}{1}$$

and projection of  $L_1$  on



$$\text{plane P is } L_2 : \frac{x}{1} = \frac{y-1}{1} = \frac{z-1}{-1}$$

vector normal to plane Q containing line  $L_1$  and  $L_2$  is

$$\vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i} + 3\hat{j}$$

Now direction Ratios of normal to plane 'P' is

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 3 & 0 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i} - \hat{j} - 6\hat{k}$$

Hence equation of plane P is

$$-3(x-0) - 3(y-1) - 6(z-1) = 0$$

$$-3x - 3y + 3 - 6z + 6 = 0$$

$$3x - 3y - 6z + 9 = 0$$

$$x + y + 2z - 3 = 0$$

Hence option (A)

Distance of plane 'P' from origin

$$\left| \frac{0+0+0-3}{\sqrt{6}} \right| = \frac{3}{\sqrt{6}} = \sqrt{\frac{3}{2}}$$

Hence (C)

Distance of plane P from (1,1,-1)

$$\left| \frac{1+1-2-3}{\sqrt{6}} \right| = \frac{3}{\sqrt{6}}$$

12. Ans. (A,B,C,D)

Since Rolle's theorem is applicable in  $[-3,3]$  therefore it is continuous and differentiable in  $[-3,3]$

Now At  $x = 1$

using continuity

$$a + b = 1 = c \quad \dots(1)$$

Also differentiable at  $x = 1$

$$\text{LHD } \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$\lim_{h \rightarrow 0} \frac{a(1-h)^2 + b - 1}{-h}$$

$$\lim_{h \rightarrow 0} \frac{a + b - 2ah + h^2 - 1}{-h}$$

$$\lim_{h \rightarrow 0} \frac{1 - 2ah + h^2 - 1}{-h} = 2a$$

Now LHD = RHD

$$2a = -1$$

$$\therefore a = -\frac{1}{2}$$

$$a + b = 1$$

$$b = \frac{3}{2}; c = 1$$

Hence  $a = -\frac{1}{2}, b = \frac{3}{2}; c = 1$

All options are correct.

### SECTION-II

1. **Ans. 9.00**

$$2A + 3BB^T = I \text{ and } B^{-1} = A^T \quad \dots(1)$$

Taking Transpose

$$(2A + 3BB^T)^T = (I)^T$$

$$2A^T + 3BB^T = I \quad \dots(2)$$

From (1) and (2)

$$A = A^T \text{ Also } A^T = B^{-1}$$

$$\therefore A = B^{-1} \text{ or } A^{-1} = B \quad \dots(3)$$

$$BA = BB^{-1} = I$$

$$BA = I \quad \dots(4)$$

$$\text{or } AB = I$$

$$\text{Again : } A^T = B^{-1}$$

$$A^TB = B^{-1}B = I$$

$$A^TB = I \quad \dots(5)$$

$$\text{From (3) } B^T = (A^{-1})^T = (A^T)^{-1}$$

$$B^T = (B^{-1})^{-1}B$$

$$\therefore B^T = B \quad \dots(6)$$

$$\text{Now } 2A + 3BB^T = I$$

$$2A + 3B^2 = I \quad (\text{from 6})$$

$$2AB + 3B^3 = B$$

$$2I + 3B^3 = B$$

$$3B^3 = B - 2I$$

$$\alpha = \det(A^{-1} - 3B^3 + 3A)$$

$$= |B - (B - 2I) + I|$$

$$\alpha = |3I| = 3^3 \quad \dots(A)$$

$$\beta = \text{tr}(A^{-1} - 3B^3 - BA)$$

$$= \text{tr}(B - (B - 2I) - I)$$

$$\beta = \text{tr}(I)$$

$$\beta = 3 \quad \dots(B)$$

$$\frac{\alpha}{\beta} = \frac{27}{3} = 9.00$$

2. **Ans. 1.00**

$$|f(x)| \leq |e^x - 1|$$

By taking limit  $x \rightarrow -\infty$  on

Both sides

$$\lim_{x \rightarrow -\infty} |f(x)| \leq |1|$$

$$\lim_{x \rightarrow -\infty} |f(x)| \leq 1$$

$\Rightarrow$  degree of  $f(x)$  cannot be  $\geq 1$

(Because if degree of  $f(x) \geq 1$  then

$$\lim_{x \rightarrow -\infty} |f(x)| = \infty$$

so  $f(x)$  must be constant function

Now for  $x = 0$

$$|f(0)| \leq |e^0 - 1|$$

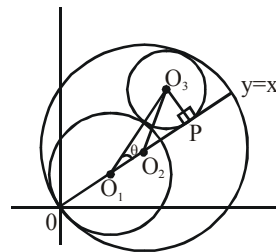
$$|f(0)| \leq 0$$

$$\therefore f(0) = 0$$

$$\text{Hence } f(x) = 0 \quad \forall x \in \mathbb{R}$$

$$f'(0) = 0$$

3. **Ans. 1.87 or 1.88**



$$S_1 = x^2 + y^2 + 3\sqrt{2}x + 3\sqrt{2}y = 0$$

$$r = 3$$

$$S_2 : x^2 + y^2 + 5\sqrt{2}x + 5\sqrt{2}y = 0$$

$$R = 5$$

Let  $O_1, O_2$  be centre of circle with radius 3 and 5 respectively.

Let  $O_3$  be the centre of circle with radius  $x$  which is tangent to two given circles and common diameter

$$O_1O_2 = 5 - 3 = 2; O_1O_3 = 3 + x,$$

$$O_2O_3 = 5 - x$$

Apply cosine rule in  $\triangle O_1O_2O_3$

$$\cos \theta = \frac{2^2 + (3+x)^2 - (5-x)^2}{2 \cdot 2 \cdot (3+x)} = \frac{4x-3}{3+x}$$

$$\text{Also in } \triangle O_1PO_3 \quad \sin \theta = \frac{x}{3+x}$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\left( \frac{4x-3}{3+x} \right)^2 + \left( \frac{x}{3+x} \right)^2 = 1$$

$$x = \frac{15}{8}$$

4. **Ans. 15.00**

We have five distinct balls and to be distributed in 3 persons. we can distribute by forming groups.

Case I : 1,1,3

$$\frac{5!}{(1!)^2 2!3!} \times 3! = 60$$

Case II : 1,2,2

$$\frac{5!}{1!(2!)^2 2!} \times 3! = 90$$

Total ways = 60 + 90 = 150

$$\frac{150}{10} = 15.00$$

5. **Ans. 0.75**

$$F(x) + F\left(x + \frac{1}{2}\right) = 1$$

$$F\left(x + \frac{1}{2}\right) + F(x+1) = 1$$

$$F(x) - F(x+1) = 0$$

$$\therefore F(x) = F(x+1)$$

$F(x)$  is periodic with period 1.

$$I = \int_0^{1500} F(x) dx = 1500 \int_0^1 F(x) dx$$

$$\Rightarrow 1500 \left[ \int_0^{\frac{1}{2}} F(x) dx + \int_{\frac{1}{2}}^1 F(x) dx \right]$$

$$\Rightarrow 1500 \left[ \int_0^{\frac{1}{2}} F(x) dx + \int_0^{\frac{1}{2}} F\left(x + \frac{1}{2}\right) dx \right]$$

$$\Rightarrow 1500 \left[ \int_0^{\frac{1}{2}} \left( F(x) + F\left(x + \frac{1}{2}\right) \right) dx \right]$$

$$1500 \left[ \int_0^{\frac{1}{2}} 1 dx \right]$$

$$1500 \times \frac{1}{2} = 750$$

$$\therefore I = \frac{750}{1000} = 0.75$$

6. **Ans. 12.56**

$$\begin{aligned} \tan^4 x + \tan^4 y + 2 \cot^2 x \cot^2 y &= 3 + \sin^2(x+y) \\ (\tan^2 x - \tan^2 y)^2 + 2 \tan^2 x \tan^2 y + 2 \cot^2 x \cot^2 y \\ &= 3 + \sin^2(x+y) \end{aligned}$$

$$\begin{aligned} (\tan^2 x - \tan^2 y)^2 + 2(\tan x \tan y - \cot x \cot y)^2 \\ + 4 &= 3 + \sin^2(x+y) \end{aligned}$$

$$\begin{aligned} (\tan^2 x - \tan^2 y)^2 + 2(\tan x \tan y - \cot x \cot y)^2 \\ + \cos^2(x+y) &= 0 \end{aligned}$$

$$\begin{aligned} \tan^2 x = \tan^2 y \text{ and } \tan x \tan y = \cot x \cot y \text{ and} \\ \cos(x+y) &= 0 \end{aligned}$$

$$\tan x = \pm \tan y ; \tan x \tan y = \pm 1, \cos(x+y) = 0$$

Case I :  $\tan x = \tan y$

$$\text{then } \tan^2 x = 1 \Rightarrow \tan x = \pm 1$$

$$\tan x = 1 \quad x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\tan x = -1 \quad x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

sum of possible values of

$$\Sigma x = 4\pi = 4 \times 3.14$$

$$\Sigma x = 12.56$$



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)  
FULL SYLLABUS  
24-01-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

### PAPER-2

#### PART-1 : PHYSICS

#### SOLUTION

##### SECTION-I

1. Ans. (A,C)

Sol.  $F = M \frac{\partial B}{\partial x} = ia^2 \times \frac{\mu_0 I}{2\pi x^2}$

Circular path  $\frac{\mu_0 I ia^2}{2\pi b^2} = \frac{mv^2}{b}$

$$k = \frac{\mu_0 I ia^2}{4\pi b}$$

for escaping  $U + k = U' + k' = 0$

$$U = -\vec{M} \cdot \vec{B}$$

$$= -\frac{\mu_0 I}{2\pi b} ia^2$$

$$k = \frac{\mu_0 I ia^2}{2\pi b}$$

2. Ans. (B,C,D)

Sol. At equator  $g'' = g - \omega^2 R$

Time period :

$$T'' = 2\pi \sqrt{\frac{L}{g''}} = 2 \text{ sec}$$

where  $g = 9.8 \text{ m/s}^2$

(A)  $g' = g > g''$  at the poles

$$T = 2\pi \sqrt{\frac{L}{g'}} = 2\pi \sqrt{\frac{L}{g}} < 2 \text{ sec}$$

(B)  $L' = L (1 + \alpha \Delta T) > L$ ;  $\alpha$  is coefficient of linear expansion

$$T' = 2\pi \sqrt{\frac{L'}{g}} > 2 \text{ sec}$$

(C)  $g''' = g_0 - \omega^2 r$  : inside satellite

$$\text{also } mg_0 = m\omega^2 r$$

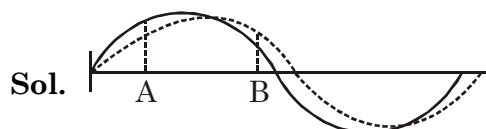
$$g''' = 0$$

$$T''' = 2\pi \sqrt{\frac{L}{g'''}} = \text{infinite} > 2 \text{ sec}$$

$$(D) \quad g_1 = \frac{g''}{\left[1 + \frac{h}{R}\right]^2} < g''$$

$$T_1 = 2\pi \sqrt{\frac{L}{g_1}} > 2 \text{ sec}$$

3. Ans. (B,C)



$$\frac{\sqrt{3}}{2} = \sin \phi_A$$

$$\phi_A = \frac{\pi}{3}$$

$$\frac{1}{2} = \sin \phi_B$$

$$\phi_B = \frac{5\pi}{6}$$

$$\frac{5\pi}{6} - \frac{\pi}{3} = \frac{2\pi}{\lambda} \times 13$$

$$\Rightarrow \frac{3}{6} = \frac{13}{\lambda} \times 2$$

$$\lambda = 52 \text{ cm}$$

$$AC = \lambda = 52 \text{ cm}$$

$$BC = 52 - 13 = 39 \text{ cm}$$

$$y = 0 \Rightarrow kx + \pi/3 = \pi \text{ \& } 2\pi$$

$$\frac{2\pi x}{52} + \frac{\pi}{3} = \pi$$

$$x = \frac{52}{3} \text{ cm \& } \frac{130}{3} \text{ cm}$$

4. Ans. (A,C)

Sol.  $\frac{dH}{dt} = \frac{4eAT_0^3(T - T_0)\sigma}{\rho V}$   
 $\propto \frac{1}{r}$

$$\frac{dH}{Adt} = 4eT_0^3(T - T_0)$$

5. Ans. (A,B)

Sol.  $v^2 = 2as = 2 \times 2 \times 200$

$$v = 20\sqrt{2} \text{ m/s}$$

$$k = \frac{1}{2} \times 1200 \times 800$$

$$= 480 \text{ kJ}$$

$$f = ma = 2400 \text{ N}$$

6. Ans. (A,C)

Sol. (1)  $f = -2.5 \text{ cm}$

$$\frac{1}{v} + \frac{1}{-25} = -\frac{1}{2.5}$$

$$\frac{1}{v} = \frac{1}{25} - \frac{10}{2.5} = -\frac{9}{25}$$

$$v = -\frac{25}{9} \text{ cm}, m = \frac{-v}{u} = -\frac{1}{9}$$

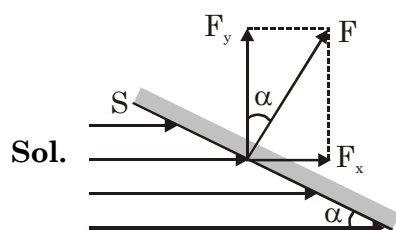
$$(2) \frac{1}{v'} - \frac{1}{25} = \frac{1}{2.5}$$

$$\frac{1}{v'} = \frac{10}{25} + \frac{1}{25} \Rightarrow v' = \frac{25}{11} \text{ cm}$$

$$m' = -\frac{v'}{u'} = +\frac{1}{11} \text{ cm}$$

### SECTION-II

1. Ans. 0.76 to 0.77



Sol.

$$\Delta p_m = 2mv \sin \alpha$$

In a unit of time the area S (figure) will be struck by all the molecules that are located at a distance v from it—that is, by molecules located in a volume  $Sv \sin \alpha$ . If n is the concentration of the molecules,

$N = nSv \sin \alpha$  molecules will strike the area S per unit time.

Thus the net momentum imparted to the object per unit time is directed perpendicular to the surface and is equal to

$$\Delta p_m \cdot N = 2mv \sin \alpha \cdot nSv \sin \alpha$$

$$= 2nmv^2 S \sin^2 \alpha,$$

$$F = 2v^2 S \sin^2 \alpha \cdot nm$$

$$F_y = F \cos \alpha = 2v^2 S \sin^2 \alpha \cos \alpha \cdot nm$$

which is maximum at  $\tan \alpha = \sqrt{2}$

$$F_{y \max} = \frac{4}{3\sqrt{3}} \rho v^2 S = 0.77 \rho v^2 S$$

2. Ans. 57.82

Sol.  $10 \text{ VSd} = 9 \text{ CSd}$

$$2 \text{ VSd} = 0.9 \text{ CSd} \times 2$$

Reading

$$= 5.5 \text{ mm} + \frac{(30 - 2 \times 0.9) \text{ CSd}}{50 \text{ CSd}} \times 0.5 \text{ mm}$$

$$= 5.5 + 0.282 \text{ mm}$$

$$= 5.782 \text{ mm}$$

3. Ans. 13.12

Sol.  $\frac{1}{f} = (1.4 - 1) \left( \frac{1}{25} - \frac{1}{-25} \right) = \frac{0.8}{25} = \frac{4}{125}$   
 $f = 31.25$

$$NS = 35 - 31.25 = 3.75 = t \left( 1 - \frac{1}{\mu} \right)$$

$$= t \left( 1 - \frac{1}{1.4} \right) = t \times \frac{0.4}{1.4}$$

$$t = 3.75 \times 3.5 = 13.125 \text{ cm}$$

$$\approx 13.12$$

4. Ans. 5.64 to 5.66

Sol.  $T = Cr^x p^y \sigma^z$

$$\Rightarrow M^0 L^0 T^1 = L^x (ML^{-3})^y (MT^{-2})^z$$

$$z = -\frac{1}{2}$$

$$y = \frac{1}{2}$$

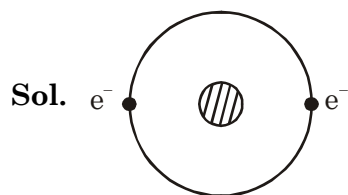
$$x = \frac{3}{2}$$

$$T = C \sqrt{\frac{\rho r^3}{\sigma}}$$

$$\frac{T}{2} = (2^3)^{1/2}$$

$$T = 4\sqrt{2} = 5.64 \text{ to } 5.66 \text{ sec.}$$

5. Ans. 61.20



Sol.

$$\frac{e^2}{4\pi\epsilon_0 r^2} - \frac{e^2}{4\pi\epsilon_0 \times 4r^2} = \frac{mv^2}{r}$$

$$\frac{3e^2}{16\pi\epsilon_0} = mv^2 r$$

$$\frac{h}{2\pi} = mvr \times 2$$

$$v = \frac{3e^2}{4\epsilon_0 h}$$

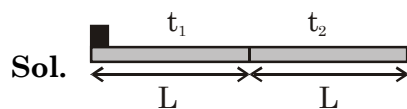
$$r = \frac{h}{4\pi \times m \times 3e^2} 4\epsilon_0 h = \frac{\epsilon_0 h^2}{3\pi m e^2}$$

$$E = \frac{3e^2}{16\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} \times 2 + \frac{e^2}{4\pi\epsilon_0 \times 2r}$$

$$= -\frac{3e^2 \times 3\pi m e^2}{16\pi\epsilon_0 \times \epsilon_0 h^2}$$

$$= 9/2 \times 13.6$$

6. Ans. 5.80 to 6.00



Sol.

$$a_b = -\mu g$$

$$2L = \frac{1}{2}(\mu g)(t_1 + t_2)^2$$

$$L = 0 \times t_2 + \frac{1}{2}\mu g t_2^2$$

$$t_2 = \sqrt{\frac{2L}{\mu g}}$$

$$t_1 = (2 - \sqrt{2})\sqrt{\frac{L}{\mu g}}$$

$$\frac{\mu mg}{2M} = a_1; a_2 = \frac{\mu mg}{M}$$

$$v_1 = \frac{\mu mg}{2M} \times t_1$$

$$v_2 = v_1 + \frac{\mu mg}{M} t_2$$

$$= \frac{\mu mg}{2M} t_1 + \frac{\mu mg}{M} t_2$$

$$\frac{v_2}{v_1} = 1 + \frac{2t_2}{t_1} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1}$$

### SECTION-III

1. Ans. 3

Sol. Time constant :  $\tau = RC$

$$C = \frac{2\pi\epsilon_0(1+K)ab}{(b-a)}$$

$$K = 2$$

$$C = \frac{6\pi\epsilon_0 ab}{(b-a)}$$

$$R = \frac{1}{2\pi\sigma} \frac{(b-a)}{(ab)}$$

$$\tau = \frac{3\epsilon_0}{\sigma} = \frac{x\epsilon_0}{\sigma}$$

$$x = 3$$

2. Ans. 3

Sol. 40 gm  $\rightarrow$  1 mole

2000 gm  $\rightarrow$  50 mole

$$1 \times C \int_T^8 \left( \frac{T}{T_0} \right)^3 dT = 50 C \int_1^T \left( \frac{T}{T_0} \right)^3 dT$$

$$\frac{8^4}{4} - \frac{T^4}{4} = \frac{50}{4} (T^4 - 1^4)$$

$$8^4 + 50 = 51T^4$$

$$T = \left( \frac{8^4 + 50}{51} \right)^{1/4} \approx 3k$$

3. Ans. 8

Sol.  $kx \cos 37^\circ = T \cos 53^\circ$

$$kx \cos 53^\circ + T \cos 37^\circ = mg$$

$$\Rightarrow \frac{T(\cos^2 53^\circ + \cos^2 37^\circ)}{\cos 37^\circ} = mg \Rightarrow T = \frac{4mg}{5}$$

$$k\vec{x} + \vec{T} + m\vec{g} = 0$$

$k\vec{x}$  does not change suddenly

$$\Rightarrow k\vec{x} + m\vec{g} = m\vec{a} = -\vec{T}$$

$$ma = \frac{4mg}{5}$$

$$\Rightarrow a = 8 \text{ m/s}^2$$

4. Ans. 0

Sol. When slider is at A, the applied potential difference across the photoelectric cell is a retarding potential difference of 10 V.

Energy of incident photon

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eVnm}}{350 \text{ nm}} = 3.5 \text{ eV}$$

Maximum KE of emitted photoelectrons is

$$KE_{\max} = \frac{hc}{\lambda} - \phi = 3.5 - 2.2 = 1.3 \text{ eV}$$

∴ Stopping potential difference

$$V_s = 1.3 \text{ V}$$

(a) The applied potential difference is

$$V (= 10 \text{ V}) > V_s (= 1.3 \text{ V})$$

Hence there is no photocurrent and reading of Ammeter is zero.

5. Ans. 1

Sol.  $\frac{dm}{dt} \times S \times \Delta T = 0.7 \text{ pJ}^2\text{V}$

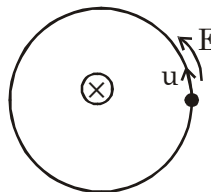
$$\frac{dm}{dt} = \frac{0.7}{10} \times 10^{-6} \times \left(\frac{3}{10^{-6}}\right)^2 \times \frac{10 \times 10^{-6}}{4200 \times 90}$$

$$= 0.01 \text{ kg/min} = 1 \times 10 \text{ gm/min}$$

6. Ans. 5

Sol.  $E = \frac{R}{2} \frac{dB}{dt} = \frac{R}{2} \alpha$

$$\frac{mdV}{dt} = qE \Rightarrow v = \frac{qE_t}{m} = \frac{qR\alpha t}{2m}$$



$$qvB \leftarrow \bullet \rightarrow N$$

$$-N + qvB = \frac{mv^2}{R}$$

$$-N = \frac{mv^2}{R} - qvB$$

$$N = \frac{-q^2\alpha^2 Rt^2}{4m} + q(B_0 + \alpha t) \frac{qR\alpha t}{2m}$$

$$= \frac{q^2\alpha^2 Rt^2}{4m} + \frac{q^2 B_0 R\alpha t}{2m}$$

$$= \frac{q^2 \alpha t R}{4m} [2B_0 + \alpha t]$$

$$= \frac{10^{-6} \times 10^{-3} \times 5 \times 1}{4 \times \frac{10^{-6}}{16}} [0.02 + 0.005]$$

$$= 5 \times 10^{-4} \text{ N}$$

## PART-2 : CHEMISTRY

## SOLUTION

### SECTION-I

1. Ans.(B,C)

2. Ans.(A,B,C)

Sol. p-chloroaniline and anilinium hydrochloride distinguished by Sandmeyer reaction  $\text{NaHCO}_3$  and  $\text{AgNO}_3$ .

Therefore option A, B and C are correct.

3. Ans.(A,B,C,D)

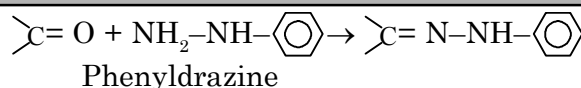
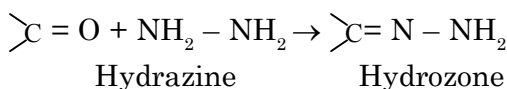
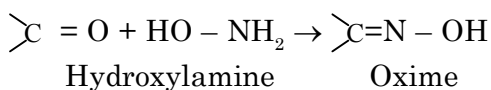
Information based on definitions

4. Ans.(A,B,D)

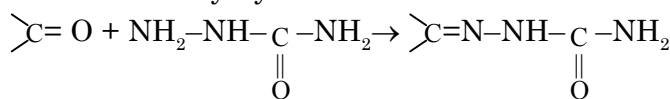
5. Ans.(A,B,C,D)

Sol. Main reaction of carbonyl compounds is Nucleophilic addition.

Reaction with ammonia derivatives are nucleophile addition Reaction.



Phenylhydrazone



Semicarbazide

Semicarbazone

Therefore option (A), (B), (C) and (D) are correct.

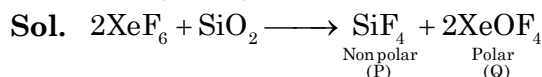
6. Ans.(A,C)

(A) conductivity will increase on adding electrolyte.

(C) Macromolecular colloids are normally lyophilic.

### SECTION-II

1. Ans.(-3.00)



$$x = 6$$

$$y = 3$$

$$\Rightarrow 3 - 6 = (-3)$$

2. Ans. (7.00)

3. (-105.50)

$$-13.7 = \left[ (\Delta_f H^0)_{\text{KCl(aq)}} + (-68.4) \right] - [(-116.5) + (-39.3)]$$

$$(\Delta_f H^0)_{\text{KCl(aq)}} = -101.1 \text{ kcal}$$

$$4.4 = (-101.1) - (\Delta_f H^0)_{\text{KCl(s)}}$$

$$(\Delta_f H^0)_{\text{KCl(s)}} = -105.5 \text{ kcal}$$

4. Ans. (4.00)

5. Ans. (4.00)

6. Ans. (94.00)

For ppt of  $\text{SrCrO}_4$

$$[\text{Sr}^{+2}] [\text{CrO}_4^{2-}] = 4 \times 10^{-9}$$

$$[\text{CrO}_4^{2-}] = 2 \times 10^{-8} \text{ M}$$

$$[\text{Ba}^{+2}] \times (2 \times 10^{-8}) = 1.2 \times 10^{-10}$$

$$[\text{Ba}^{+2}] = 0.6 \times 10^{-2} \text{ M}$$

$$\% \text{ of } \text{Ba}^{2+} \text{ ppt} = \frac{(0.1 - 0.6 \times 10^{-2})}{0.1} \times 100 = 94\%$$

### SECTION-III

1. Ans. (2)

2. Ans. (4)

3. Ans. (3)

$$V = \frac{58.5}{2.167} \approx 27 \text{ cm}^3$$

$$a^3 = 27 \text{ cm}^3$$

$$a = 3 \text{ cm}$$

4. Ans. (9)

5. Ans. (4)

6. Ans. (3)

$$\text{CH}_4 \quad \text{G}$$

$$\text{V} \quad \text{V}$$

$$\text{n} \quad \text{n}$$

$$\text{T} \quad \text{T}$$

$$\sigma \quad \sigma$$

$$\frac{(u_{\text{avg}})_{\text{CH}_4}}{(u_{\text{rms}})_{\text{G}}} = \frac{\sqrt{\frac{8}{\pi \times 16}}}{\sqrt{\frac{3}{M_{\text{G}}}}}$$

$$\frac{48}{\pi} = \frac{8}{16\pi} \times \frac{M_{\text{G}}}{3}$$

$$M_{\text{G}} = 288$$

$$\frac{(Z_1)_{\text{CH}_4}}{(Z_1)_{\text{G}}} = \frac{\sqrt{\frac{1}{16}}}{\sqrt{2} \times \sqrt{\frac{1}{288}}} = \sqrt{\frac{288}{32}} = \sqrt{\frac{9}{1}} = \frac{3}{1}$$

## PART-3 : MATHEMATICS

## SOLUTION

### SECTION-I

1. Ans. (A,C)

$(1 + \lambda) > 0$ , graph of

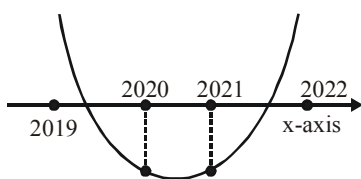
$y = (x - 2019)(x - 2022) + \lambda(x - 2020)(x - 2021)$  is

$$f(2019) = 2\lambda$$

$$f(2020) = -2$$

$$f(2021) = -2$$

$$f(2022) = 2\lambda$$



2. Ans. (B,D)

$y = m(x - a)$  is a focal chord so, angle between normals drawn at A and B =  $90^\circ$   
length of focal chord =  $4a \operatorname{cosec}^2 \alpha$

$$\text{for } m = 1, \alpha = \frac{\pi}{4} \Rightarrow \operatorname{cosec}^2 \frac{\pi}{4} = 2$$

$$AB = 4a \times 2 = 8a$$

3. Ans. (A,C)

$n(A)$  = number of elements in A = m

$n(B)$  = number of elements in B =  $12 - m$

$m \neq 0, 6, 12$

For A, we have 10 elements to choose from

$$N = \sum_{m=1}^{11} {}^{10}C_{m-1} - {}^{10}C_5 = 272$$

4. Ans. (B,C)

$$(Z_1 - Z_0)(\bar{Z}_1 - \bar{Z}_0) = 16R^2$$

$$\Rightarrow Z_1 \bar{Z}_1 - Z_0 \bar{Z}_1 - \bar{Z}_0 Z_1 + |Z_0|^2 = 16R^2 \quad \dots(1)$$

$$\left( \frac{1}{\bar{Z}_1} - Z_0 \right) \left( \frac{1}{Z_1} - \bar{Z}_0 \right) = 256R^2$$

$$\frac{1}{Z_1 \bar{Z}_1} - \frac{Z_0}{Z_1} - \frac{\bar{Z}_0}{\bar{Z}_1} + |Z_0|^2 = 256R^2$$

$$\text{put } |Z_1|^2 = k$$

$$1 - Z_0 \bar{Z}_1 - \bar{Z}_0 Z_1 + K |Z_0|^2 = 256R^2 K$$

$$(1) - (2)$$

$$\Rightarrow (K - 1) + (1 - K) |Z_0|^2 = 16R^2(1 - 16K)$$

$$\Rightarrow (1 - K) + (|Z_0|^2 - 1) = 16R^2(1 - 16K)$$

$$\Rightarrow (1 - K) = (1 - 16K)$$

$$\Rightarrow K = \frac{31}{511}$$

5. Ans. (A,D)

$$A(0,0,0), B(1,2,3)$$

one two points on  $L_1$

foot of perpendicular of B in  $P_1$

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1} = -\frac{(1+2+3)}{3} = -2$$

$$\Rightarrow x = -1, y = 0, z = 1$$

$$Q(-1,0,1)$$

$$D.R^5 \text{ of } AQ < -1, 0, 1 >$$

$$a : b : c :: -1, 0, 1$$

6. Ans. (A,B)

$$P''(x) = 6(x-1)$$

$$\Rightarrow P'(x) = \frac{6(x-1)^2}{2} + C_1$$

$$0 = 3 + C_1 \Rightarrow C_1 = -3$$

$$P'(x) = 3(x-1)^2 - 3$$

$$P(x) = (x-1)^3 - 3x + C$$

$$0 = -3 + C$$

$$P(x) = (x-1)^3 - 3x + 3$$

$$= (x-1)^3 - 3(x-1)$$

$$= (x-1) [x^2 - 2x - 2]$$

### SECTION-II

1. Ans. 4.80

$$\text{We have } A^3 - 6A^2 + 7A + 2I = 0$$

$$\text{we have } A^3 - 6A^2 + 7A + 2I = 0$$

$$\Rightarrow B = 2I$$

$$|B| = 8 \Rightarrow |B|^{3-2} = 8$$

$$\Rightarrow C = 8B = 16I$$

2. Ans. 0.50

$$N = \int_0^1 x \left\{ \frac{d}{dx} \left( 4 \times (1-x^2)^3 \times (-2x) \right) \right\} dx$$

$$= -8x^2 (1-x^2)^3 \int_0^1 + 8 \int_0^1 (1-x^2)^3 \times x dx$$

$$= 0 + 4 \int_0^1 (1-x^2)^3 (2x) dx$$

$$= -4 \int_1^0 t^3 dt = 1$$

3. Ans. 0.84

$$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$$

$$\Rightarrow (\vec{r} - \vec{c}) \times \vec{b} = \vec{0}$$

$$\vec{r} = \vec{c} + \lambda \vec{b}$$

$$0 = \vec{c} \cdot \vec{a} + \lambda \vec{b} \cdot \vec{a}$$

$$= 1 + 2 + 3 + (3 + 4 + 3)\lambda$$

$$\lambda = -\frac{6}{10} = -\frac{3}{5}$$

$$\vec{r} = \hat{i} + \hat{j} + \hat{k} - \frac{9}{5}\hat{i} - \frac{6}{5}\hat{j} - \frac{3}{5}\hat{k}$$

$$\vec{r} = -\frac{4}{5}\hat{i} + \frac{-1}{5}\hat{j} + \frac{2}{5}\hat{k}$$

$$|\vec{r}|^2 = \frac{21}{25} = 0.84$$

4. Ans. 0.50

point of intersection of

$$y = (2 - \sqrt{3})x \text{ and } y - 1 = x(2 + \sqrt{3}) + 2 + \sqrt{3}$$

$$\Rightarrow (2 - \sqrt{3})x - 1 = x(2 + \sqrt{3}) + 2 + \sqrt{3}$$

$$\Rightarrow -2\sqrt{3}x = 3 + \sqrt{3} \Rightarrow x = \frac{\sqrt{3} + 1}{-2}$$

$$y = (2 - \sqrt{3}) \frac{(1 + \sqrt{3})}{(-2)} = \frac{2 + \sqrt{3} - 3}{-2} = \frac{-1 + \sqrt{3}}{-2}$$

$$\Rightarrow \alpha = \frac{\sqrt{3} - 1}{2}, \beta = \frac{\sqrt{3} + 1}{2}$$

$$\Rightarrow \alpha\beta = \frac{2}{4} = \frac{1}{2} = 0.50$$

5. Ans. 0.24 or 0.25

$$P\left(\frac{W}{L}\right) = \frac{2}{3}, P\left(\frac{L}{W}\right) = \frac{1}{3}$$

$$P\left(\frac{W}{L}\right) = \frac{1}{3}, P\left(\frac{L}{L}\right) = \frac{2}{3}$$

Required probability

$$= P(WWWL) + P(WWLW) + P(WLWW) + P(LWWW)$$

$$= \left(\frac{2}{3}\right)^3 \times \frac{1}{3} + \left(\frac{2}{3}\right)^2 \times \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^2 \times \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^2 \times \left(\frac{2}{3}\right)^2$$

$$= \frac{8 + 4 + 4 + 4}{3^4} = \frac{20}{81} = .24, .25$$

6. Ans. 1.50

$$\frac{T_{n+1}}{2^{n+1}} = \frac{T_n}{2^n} + \frac{1}{2} \times 2^n$$

$$\Rightarrow \frac{T_2}{2^2} - \frac{T_1}{2} = \frac{1}{2} \times 2$$

$$\frac{T_3}{2^3} - \frac{T_2}{2^2} = \frac{1}{2} \times 2^2$$

$$\frac{T_n}{2^n} - \frac{T_{n-1}}{2^{n-1}} = \frac{1}{2} \times 2^{n-1}$$

$$\frac{T_n}{2^n} - \frac{T_1}{2} = \frac{1}{2} \frac{(2^{n-1} - 1) \times 2}{2^{n-1}} = 2^{n-1} - 1$$

$$\Rightarrow T_n = 2^{2n-1} - 2^{n-1}$$

$$\Rightarrow S_n = \frac{2 \cdot 4^n - 2 - 3 \cdot 2^n + 3}{3}$$

$$= \frac{2 \cdot 4^n - 3 \cdot 2^n + 1}{3}$$

### SECTION-III

1. Ans. 1

$$D = \begin{vmatrix} 1 & 1 & -2 \\ 2 & -3 & 1 \\ 1 & -5 & 4 \end{vmatrix} = 0$$

and system has solution

$$\Rightarrow \begin{vmatrix} 0 & 1 & -2 \\ 0 & -3 & 1 \\ k & -5 & 4 \end{vmatrix} = 0 \Rightarrow k = 0$$

on solving given equations

$$\Rightarrow x = y = z = \lambda (\text{suppose})$$

$$\text{As } x_0, y_0, z_0 \in \mathbb{R} - \{0\}, \lambda \neq 0$$

$$\frac{x_0 y_0 + y_0 z_0 + z_0 x_0}{x_0^2 + y_0^2 + z_0^2} = \frac{\lambda^2 + \lambda^2 + \lambda^2}{\lambda^2 + \lambda^2 + \lambda^2} = 1$$

2. Ans. 2

$$f(x) = ax^n + bx^{n-1} + cx^{n-2} + \dots, n \geq 2$$

$$2 + 2(ax^n + bx^{n-1} + cx^{n-2} + \dots)$$

$$= a\{(x+1)^n + (x-1)^n\} + b\{(x+1)^{n-1} + (x-1)^{n-1}\} + c\{(x+1)^{n-2} + (x-1)^{n-2}\}$$

+ ...

compare co-eff. of  $x^{n-2}$

$$\Rightarrow 2c = a \times {}^nC_2 \times 2 + 2c$$

$$\Rightarrow 0 = an(n-1) \text{ not possible}$$

similarly not possible for  $n < 2$

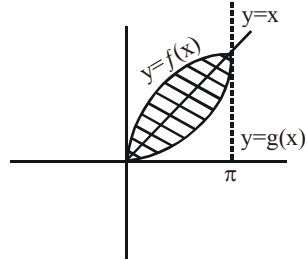
for  $n = 2$   $f(x) = x^2 + bx + c$  satisfies

3. Ans. 1

$$(2x + 3y - 8)(x + 2y - 5) = 40$$

$$\Rightarrow a = 2, b = 6, h = \frac{7}{2}, g = -9, f = -\frac{31}{2}$$

4. Ans. 4



$$A = \int_0^\pi (f(x) - g(x)) dx = 2 \int_0^\pi (f(x) - x) dx = 4$$

5. Ans. 2

Use  $\tan \theta = \cot \theta - 2 \cot 2\theta$

$$\Rightarrow \tan x + \frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \left( \frac{x}{2^2} \right) + \dots$$

$$\dots + \frac{1}{2^{n-1}} \tan \left( \frac{x}{2^{n-1}} \right) = \frac{1}{2^{n-1}} \cot \frac{x}{2^{n-1}} - 2 \cot 2x$$

Now differentiate w.r.t  $x$

$$\Rightarrow \sec^2 x + \frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \dots$$

$$+ \frac{1}{2^{n-2}} \sec^2 \frac{x}{2^{n-1}}$$

$$= 4 \operatorname{cosec}^2 2x - \frac{1}{2^{n-2}} \operatorname{cosec}^2 \frac{x}{2^{n-1}}$$

$$\Rightarrow a = 2$$

6. Ans. 3

Replace  $x$  by  $\frac{x-1}{x}$ , we get

$$f\left(\frac{x-1}{x}\right) + f\left(\frac{1}{1-x}\right) = \tan^{-1}\left(\frac{x-1}{x}\right) \dots (2)$$

Replace  $x$  by  $\frac{1}{1-x}$  in (1) we get

$$f\left(\frac{1}{1-x}\right) + f(x) = \tan^{-1}\left(\frac{1}{1-x}\right) \dots (3)$$

(1) - (2) + (3), we get

$$2f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{1-x}\right) - \tan^{-1}\left(\frac{x-1}{x}\right) \dots (4)$$

Also,

$$2f(1-x)$$

$$= \tan^{-1}(1-x) + \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{x}{x-1}\right) \dots (5)$$

put  $x = \frac{1}{2}$  and add (4) and (5)

$$\Rightarrow 4f\left(\frac{1}{2}\right) = \frac{3\pi}{2}$$

$$\Rightarrow N = f\left(\frac{1}{2}\right) = \frac{3\pi}{8}$$